

DTC	P1520	Stop Light Switch Signal Malfunction (Only for A/T)
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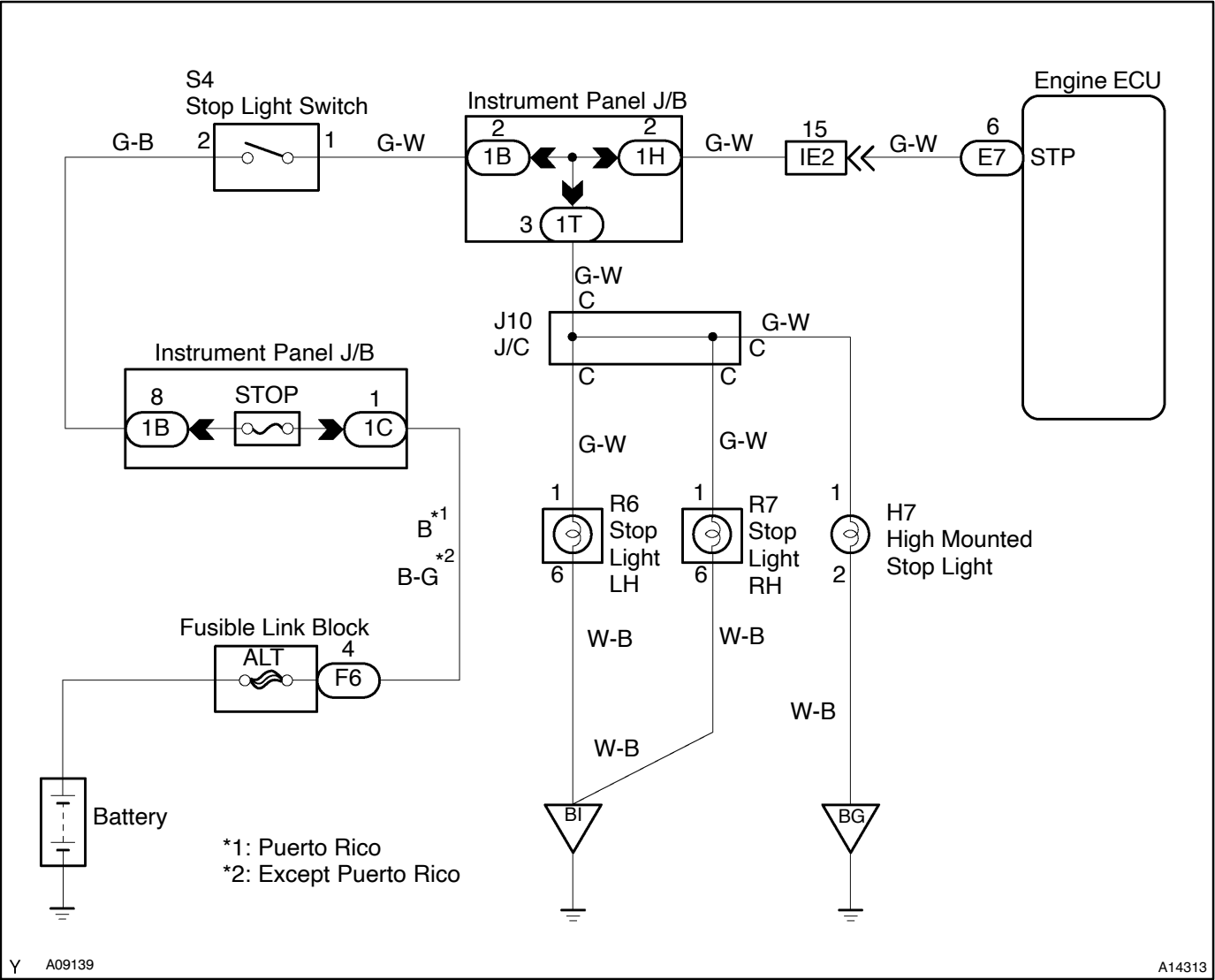
CIRCUIT DESCRIPTION

This signal is used to detect when the brakes have been applied. The STP signal voltage is the same as the voltage supplied to the stop lights.

The STP signal is used mainly to control the fuel cut-off engine speed (The fuel cut-off engine speed is reduced slightly when the vehicle is braking.).

DTC No.	DTC Detecting Condition	Trouble Area
P1520	The stop light switch does not turn off even once the vehicle is driven (2 trip detection logic)	• Short in stop light switch signal circuit • Stop light switch • ECM

WIRING DIAGRAM



INSPECTION PROCEDURE

HINT:

Read freeze frame data using TOYOTA hand-held tester or OBD II scan tool. Because freeze frame records the engine conditions when the malfunction is detected, when troubleshooting it is useful for determining whether the vehicle was running or stopped, the engine warmed up or not, the air-fuel ratio lean or rich, etc. at the time of the malfunction.

1 Check operation of stop light.

PREPARATION:

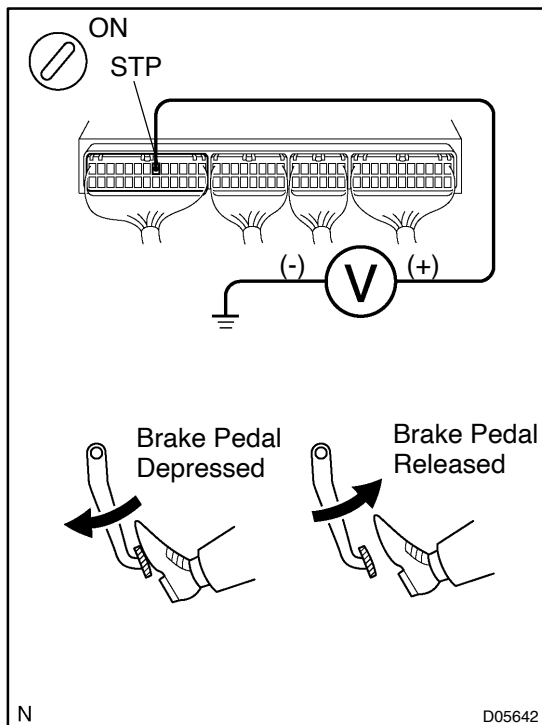
Check if the stop lights go on and off normally when the brake pedal is operated and released.

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Check and repair stop light circuit (See page [BE-37](#)).

OK

2 Check STP signal.



When using TOYOTA hand-held tester:

PREPARATION:

- Connect the to the DLC3.
- Turn the ignition switch ON and TOYOTA hand-held tester main switch ON.

CHECK:

Read the STP signal on the TOYOTA hand-held tester.

OK:

Brake pedal is depressed: STP ... ON

Brake pedal is released: STP ... OFF

When not using TOYOTA hand-held tester:

PREPARATION:

- Remove the connector cover from the ECM.
- Turn the ignition switch ON.

CHECK:

Check voltage between terminal STP of ECM and body ground.

OK:

Brake pedal	Voltage
Depressed	7.5 - 14 V
Released	Below 1.5 V

OK

Check for intermittent problems (See page [DI-3](#)).

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3	Check harness and connector between ECM and stop light switch (See page IN-29).
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Repair or replace harness or connector.

OK

Check and replace ECM (See page [IN-29](#)).